

SI CONVERSION FACTORS

This table lists units of measurements from non-SI systems and the factors needed to convert them to SI units. The conversion factor in the column headed "SI equivalent" can be used to convert from the non-SI unit to the SI unit named. The reverse conversion can be made using the conversion factor in the column headed reciprocal.

UNIT	SYMBOL	QUANTITY	SI EQUIVALENT	SI UNIT	RECIPROCAL
Acre		area	0.405	hm ²	2.471
Ångström	Å	length	0.1	nm	10
Astronomical unit	AU	length	0.150	Tm	6.684
Atomic mass unit	amu	mass	1.661 × 10 ⁻²⁷	kg	6.022 × 10 ²⁶
Bar	bar	pressure	0.1	MPa	10
Barrel (US) = 42 US gal	bbl	volume	0.159	m ³	6.290
Calorie	cal	energy	4.187	J	0.239
Cubic foot	cu ft	volume	0.028	m ³	35.315
Cubic inch	cu in	volume	16.387	cm ³	0.061
Cubic yard	cu yd	volume	0.765	m ³	1.308
Curie	Ci	activity of radionuclide	37	GBq	0.027
Degree Celsius	°C	temperature	1	K	1
Degree Fahrenheit	°F	temperature	0.556	K	1.8
Electronvolt	eV	energy	0.160	aJ	6.241
Erg	erg	energy	0.1	μJ	10
Fathom (6 ft)		length	1.829	m	0.547
Fermi	fm	length	1	fm	1
Foot	ft	length	30.48	cm	0.033
Foot per second	ft s ⁻¹	velocity	0.305 1.097	m s ⁻¹ km h ⁻¹	3.281 0.911
Gallon (UK)	gal	volume	4.546	dm ³	0.220
Gallon (US) = 231 cu in	gal	volume	3.785	dm ³	0.264
Gauss	Gs, G	magnetic flux density	100	μT	0.01
Grain	gr	mass	1	g	15.432
Hectare	ha	area	0.746	hm ²	1
Horsepower	hp	power	2.54	kW	1.341
Inch	in	length	9.807	cm	0.394
Kilogram-force	kgf	force	1.852	N	0.102

UNIT	SYMBOL	QUANTITY	SI EQUIVALENT	SI UNIT	RECIPROCAL
Knot	kn	velocity	9.461 × 10 ¹⁵	km h ⁻¹	0.540
Light-year	ly	length	1	m	1.057 × 10 ⁻¹⁶
Liter	l	volume	1,193.3	dm ³	1
Mach number	Ma	velocity	10	km h ⁻¹	8.380 × 10 ⁻⁴
Maxwell	Mx	magnetic flux	1	nWb	0.1
Micron	μ	length	1.852	μm	1
Mile (nautical)		length	1.609	km	0.540
Mile (statute)		length	1.609	km	0.621
Miles per hour (mph)	mile h ⁻¹	velocity	2.91 × 10 ⁻⁴	km h ⁻¹	0.621
Ounce (avoirdupois)	oz	mass	31.103	g	0.035
Ounce (troy) = 480 gr		mass	30,857	g	0.032
Parsec	pc	length	10	Tm	0.0000324
Pint (UK)	pt	volume	0.1	dm ³	1.760
Pound	lb	mass	4.448	kg	2.205
Pound force	lbf	force	6.895	N	0.225
Pound force/in		pressure	0.138	kPa	0.145
Pounds per square inch	psi	pressure	0.01	kPa	0.145
Röntgen	R	exposure	0.258	mC kg ⁻¹	3.876
Second = (1/60')	"	plane angle	4.85 × 10 ⁻⁶	mrاد	2.063 × 10 ⁵
Solar mass	M	mass	1.989 × 10 ³⁰	kg	5.028 × 10 ⁻³¹
Square foot	sq ft	area	9.290	dm ²	0.108
Square inch	sq in	area	6.452	cm ²	0.155
Square mile (statute)	sq mi	area	2.590	km ²	0.386
Square yard	sq yd	area	0.836	m ²	1.196
Stere	st	volume	1	m ³	1
Therm = 105 btu		energy	0.105	GJ	9.478
Ton = 2240 lb		mass	1.016	Mg	0.984
Ton-force	tonf	force	9.964	kN	0.100
Ton-force/sq in		pressure	15.444	MPa	0.065
Tonne	t	mass	1	Mg	1